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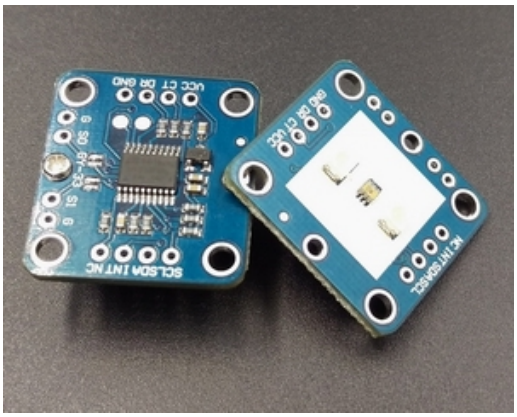
GY-33Color Sensor Module User ManualV1.0

I. Overview

GY-33It is a low-cost color recognition sensor module.



Operating voltage3-5VIt has low power consumption, small size, and is easy to install. Its working principle is lighting...ledLight is emitted and irradiated onto the test object.

After the object, the reflected light is detected by the filter.RGBProportion ,according toRGBThe proportion value is used to identify the color.

This module has two ways to read data, namely serial portUART(TTL(Level) orIIC(2(Line). The baud rate of the serial port is... 9600bpsand115200bpsIt is configurable and has two output modes: continuous and query. It can save settings even after power failure.

There are simple ones7Color recognition, no calculation requiredRGBValue. It can adapt to different working environments and connect to microcontrollers and computers.

The module can also be configured to operate in a separate sensor chip mode.

As a simple sensor moduleMCUThey do not participate in data processing.

supplyarduino,51,stm32This is a microcontroller communication program; schematics and internal microcontroller source code are not provided.

II. Product Features

(1High cost performance

(2),built-inMCUCalculate color (3), IICSerial communication format (4 Adjustableledbrightness

(5), and provide corresponding host computer software

III. Product Application

(1), intelligent robots

(2Teaching laboratory equipment

(3Production line product testing

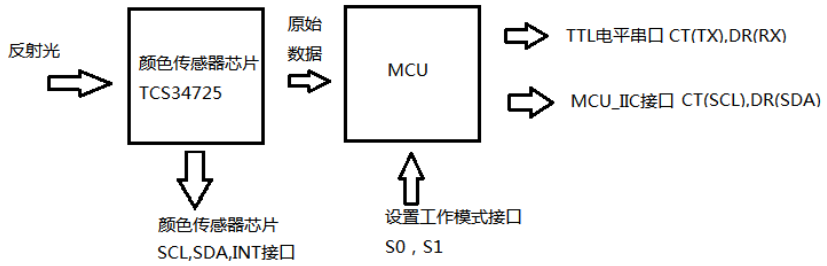
(4), colorimeter

(5),ledColor detection

Technical parameters

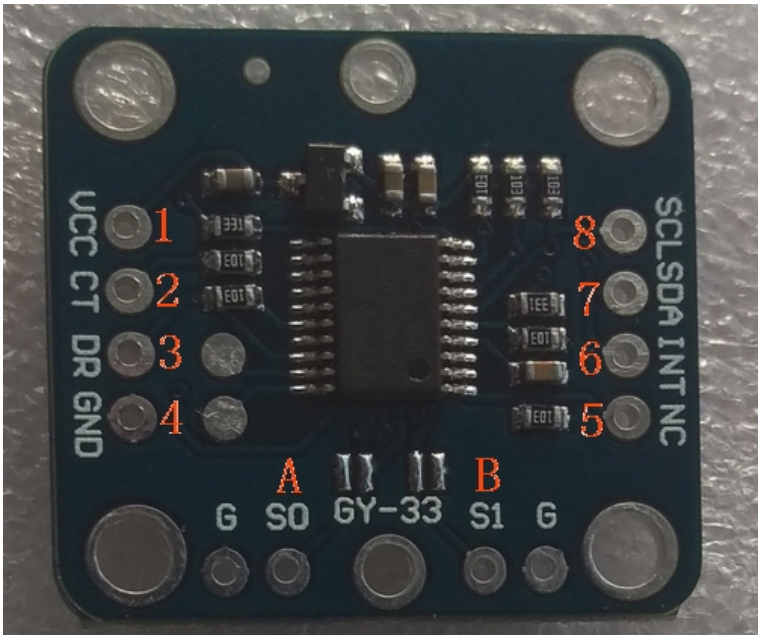
name	parameter
Measurement range	RGB 0-255
Response frequency	10 Hz
Operating voltage	3~5V
Operating current	15mA
Operating temperature	- 20°~85°
Storage temperature	- 40°~125°
size	24.3mm×26.7mm
sensor chip	TCS34725

Interface block diagram:



III. Pin Description

Actual product photos (labeled):



Pin1	VCC	Power supply + (3V-5V)
Pin2	CT	serial portUART_TX / IIC_SCL
Pin3	DR	serial portUART_RX / IIC_SDA
Pin4	GND	power ground
Pin5	NC	Keep, do not connect
Pin6	INT	TCS34725Color chip interruptedS1=0(catchGND(Activated at time)
Pin7	SDA	TCS34725Color chip data cableS1=0(catchGND(Activated at time)
Pin8	SCL	TCS34725Color chip clock lineS1=0(catchGND(Activated at time)
PinA	S0	Serial port/MCU_IICMode Selection
PinB	S1	Select using only sensor chip

Note: ①,PinA(S0Hardware selection module operating mode,Pin2(CT),Pin3(DR)forGY-33Module communication interface,

S0=1(default)	serial portUARTmodel,Pin2forTX, Pin3forRX, TTLLevel
S0=0(catchGNDhour)	MCU_IICmodel,Pin2forSCL, Pin3forSDA,

② ,PinB(S1Using only sensor chipsTCS34725Mode, SelectMCUWhether or not they participate in data processing work,

S1=1(default)	MCU + TCS34725model,Pin7, Pin8Please do not create any links.
S1=0(catchGNDhour)	onlyTCS34725model,Pin7For chipsSCL, Pin8For chipsSDA

V. Communication Protocol

❶ **Serial port protocol:**whenGY-33Module hardwarePinA(S0）=1When to use

(1Serial communication parameters (default baud rate value)9600bps(Can be set via software)

Baud rate:9600 bps	Check digit:N	Data bits:8	Stop bit:1
Baud rate:115200 bps	Check digit:N	Data bits:8	Stop bit:1

(2), module output format, each frame contains8-13Bytes (hexadecimal):

① Byte0:	0x5A	Frame header flags
②.Byte1:	0x5A	Frame header flags
③.Byte2:	0X25	Data type of this frame (see explanation for reference)
④.Byte3:	0x06	Data volume (hereinafter)3(Taking one data point as an example)
⑤.Byte4:	0x00~0xFF	data1high8Bit
⑤.Byte5:	0x00~0xFF	data1Low8Bit
⑥.Byte6:	0x00~0xFF	data2high8Bit
⑦.Byte7:	0x00~0xFF	data2Low8Bit
⑧.Byte8:	0x00~0xFF	data3high8Bit
9. Byte9:	0x00~0xFF	data3Low8Bit
⑩.Byte10:	0x00~0xFF	Checksum (the sum of all preceding data, keeping only the lowest value)8Bit)

Byte2Explanation of the meaning of "represented":

Byte2	0x15	0x25	0x45	0x55
meaning:	originalRGBC	Brightness, color temperature, simple colors	MCUAfter processingRGB	IICaddress

(3), data calculation methods

① OriginalRGBCCalculation method (when)Byte2 = 0x15hour) :

R: Red=(high8Position <<8)|Low8The bit represents the value after passing through the red filter.ADvalue

G: Green=(high8Position <<8)|Low8The bit representation after passing through the green filterADvalue B:

Blue=(high8Position <<8)|Low8The bit representation after passing through the blue filterADvalue C:

Clear=(high8Position <<8)|Low8The bit indicates that there is no filter.ADExample: One frame of data

<5A-5A-15-08-01-78-01-92-00-4C-05-05-33 >

R=(0x01<<8)|0x78=376

G=(0x01<<8)|0x92=402

B=(0x00<<8)|0x4c=76

C = (0x05 << 8) | 0x05 = 1285

② Brightness, color temperature, and simple colors, calculation methods (when...)Byte2 = 0x25(Time):

Brightness:Lux=(high8Position <<8)|Low8Unit of measurementlux Color temperature:CT=(high8Position <<8)|

Low8Unit of measurementK(Kelvin) Simple7The colors are shown in the table below:

color	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
color	blue	Dark blue	green	black	White	pink	yellow	red

Bit7-bit0:Which location1Indicates the color of this bit.

color:color=(high8Position <<8)|Low8Bit;

Example: A frame of data

< 5A-5A-25-06-02-CC-0C-5D-00-02-18 >

Lux=(0x02<<8)|0xcc=716 (lux)

CT=(0x0c<<8)|0x5d=3162(K)

Color=(0x00<<8)|0x02=2((Currently yellow)

③ After processingRGBValue calculation method (when)Byte2=0x45 hour): R: R = Byte4; expressRAfter processing8bitvalue G:G = Byte5; expressGAfter processing8bitvalue B:B = Byte6; expressBAfter processing8bitExample: One frame of data

<5A-5A-45-03-FF-FF-4C-46>

R=FF=255;

G=FF=255;

B=4C=76;

④IICaddress(whenByte2 = 0x55(Time): Returned1indivual8bit

The data is the current one.IICaddress

(4Command bytes, sent by the external controller toGY-33Module (hexadecimal)

1All serial port command formats, frame header:0xa5

Command format: Frame header + command + checksum8bit)

2Serial port commands:

① Serial port output configuration register:

command	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
Output command	Auto	0	0	0	0	RGBC	LCC	RGB
Auto(default1)	1:After powering on, the output will be configured according to the previous output settings.0It does not output automatically after power-on.							
bit6-bit3	Requires being set to zero:0000							
RGBC(default0)	1:Continuous outputRGBCoriginalADvalue,0:No output; whenAutoPlace1Power off save							
LCC(default0)	1:Continuous output of brightness, color temperature, and color value.0:No output; whenAutoPlace1Power off save							
RGB(default1)	1:After continuous output processing8-bit RGBvalue,0:No output; whenAutoPlace1Power off save							

Command format:0xA5+command+sum

example:bit7 (Auto=1), bit0 (RGB =1)

Send command:0xA5+0x81+0x26, indicating continuous outputMCUAfter processingRGBThis setting is saved after a power outage and will automatically continue output after power is restored.MCUAfter processingRGBvalue;

②,ledBrightness configuration instructions:

0xA5+0x6X+sum----------Function byte0x6XmiddleXCan be taken0arrive10(Right now0x60-0x6A),XThe smaller the value, ledThe brighter,sum equal0xA5+0x6XThe lower of the sum8Position, or via host computer "ledBrightness settings scroll bar settings, total11Each level (this setting)The default values are not saved after power failure and are retained upon power-on.0x63)

0xA5+0xCC+0x71 ----------Save currentledBrightness setting command (supported in the latest version, saved after power failure, takes effect after power-on; can be saved to the module when the object being tested emits its own light).led(It's never lit)

③ Query output command:

0xA5+0x51+0xF6 ----------OutputRGBCoriginalADValue (the module returns a data type of)0x15)

0xA5+0x52+0xF7 ----------Output brightness, color temperature, and color value (the module returns a data type of...).0x25) 0xA5+0x54+0xF9 --------- -----Output after processing8-bit RGBValue (the module returns a data type of)0x45) 0xAA+0xF5+0x9F----------Output moduleIICaddress (The module returns a data type of)0x55)

Note: If using query output, please ensure that you have configured it beforehand.command=0x00

④ Baud rate configuration:

0xA5+0xAE+0x53 ----------9600(default)

0xA5+0xAF+0x54 ----------115200

⑤ White balance calibration instructions:

0xA5+0xBB+0x60----------Used during color white balance calibration (this command saves calibration results even when power is off). When adjusting the illuminance...led

When adjusting brightness, white balance calibration is required. The method is as follows: find a relatively white object, get close to the object, and then send a white balance calibration command via serial port.

⑥, SettingsIICAddress command: (This command saves the modified version even after power failure)IICaddress) 0xAA+XX+sum---------

XXexpress7bit IICaddress,sumequal0xAA+XXThe lower of the sum8For example, the original factoryIIC 7bitAddress is0x5A,Then send0xAA+0x5A+04To the module,7bitShift the address left by one bit, and the module... 8bit IICAddress is0xB4

⑦ Change the measurement range (select a shorter time when the ambient light is high; the latest version supports this; the changes are saved after power failure and take effect after power is restored). 0xA5+0x58+0xFD ----------T=700msThe data output interval is approximately equal toT 0xA5+0x59+0xFE----------T=154msThe data output interval is approximately equal toT 0xA5+0x5A+0xFF ---------- T=100msThe data output interval is approximately equal toT(default) 0xA5+0x5B+0x00 ---------- T=24msThe data output interval is approximately equal toT 0xA5+0x5C+0x01 ---------- -T=2.4msThe data output interval is approximately equal to100ms

TThe time it takes for the module to collect colors,TThe larger,RGBCA higher value indicates a better fit for darker environments; conversely, a lower value indicates a better fit for darker environments.TThe smaller, RGBCThe smaller the value, the more suitable it is for brighter environments, such as detection.ledColor, etc.

❷, MCUIICprotocol:whenGY-33Module hardwarePinA(S0）=0When to use

① IICAddress, default7bitAddress is0x5A,but8bitAddress is0xB4

IICThe address can be modified via serial port configuration.128Different addresses are saved even after power failure.

② IICregister:

0x00(Read-only)	RAW_RED_H	originalRed ADHigh value8Bit
0x01(Read-only)	RAW_RED_L	originalRed ADLow value8Bit
0x02(Read-only)	RAW_GREEN_H	originalGreen ADHigh value8Bit
0x03(Read-only)	RAW_GREEN_L	originalGreen ADLow value8Bit
0x04(Read-only)	RAW_BLUE_H	originalBlue ADHigh value8Bit
0x05(Read-only)	RAW_BLUE_L	originalBlue ADLow value8Bit
0x06(Read-only)	RAW_CLEAR_H	originalClear ADHigh value8Bit
0x07(Read-only)	RAW_CLEAR_L	originalClear ADLow value8Bit
0x08(Read-only)	Lux_H	High brightness8Bit
0x09(Read-only)	Lux_L	Low brightness8Bit
0x0a(Read-only)	CT_H	High color temperature8Bit

0x0b(Read-only)	CT_L	Low color temperature8Bit
0x0c(Read-only)	R	After processing8bit rvalue
0x0d(Read-only)	G	After processing8bit gvalue
0x0e(Read-only)	B	After processing8bit bvalue
0x0f(Read-only)	Color	Simple color status
0x10(Reading and writing)	Config	Configuration Register

Register description:

Simple Color Status Register0x0f:

	Color	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
0x0f	color	blue	Dark blue	green	black	White	pink	yellow	red

Note:Bit7-bit0:Which location1Indicates the color of that bit, for exampleColor=0x02The current object is yellow.

Configuration Register0x10:

	Config	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
0x10	Configuration	D4	D3	D2	D1	NC	NC	NC	White Balance

bit7-bit4	0-10Eleven levels of adjustmentledThe brightness level is set to [default level].3
bit3-bit1	reserve0000
bit0	1White balance calibration;0Do not perform white balance

VI. Module Usage Method

Module data description: Raw RGBC: Raw AD data, without any processing;

Brightness, color temperature, simple color, and processed RGB values: obtained through the corresponding algorithm of the MCU;

The module data update frequency is approximately 10 Hz;

This module is a serial port and I2C output module; the module defaults to...Serial port mode.

Serial port mode (default): Module (Pin A (S0) on the leftWhen disconnected, it is in serial port mode.

The accompanying host computer allows for convenient configuration of the module. The host computer's command prompt corresponds to the module's serial port.

In the command register, set the corresponding position to 1, then click the "Send" button in the lower right corner to configure the module.

The block will send out the corresponding data, which will be displayed in the corresponding box on the left. For example, to perform white balance calibration, you can do so in "Open".

Enter the corresponding command below the "Serial Port" button, and then click the send button on the right. Changing the baud rate is the same; [the text abruptly ends here, likely due to an incomplete sentence or missing information.]

Press the LED brightness adjustment button (triangle button) to change the LED brightness. The brightness setting level is displayed in the left-hand box; the higher the value, the lower the brightness.

The larger the LED, the dimmer it is.

MCU_IIC communication mode: connect the module (Pin A (S0) on the leftConnecting to GND, the module enters IIC mode, and IIC communication is established.

The clock signal needs to be between 40kHz and 200kHz, which can be achieved by reading the corresponding registers. The interval between all data reads should be less than 10Hz.

IIC supports "LED brightness adjustment" in module configuration. "White balance calibration".

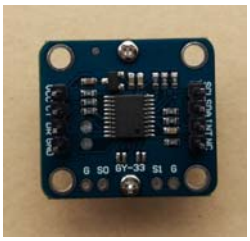
Sensor chip only mode: Module(PinB on the right, S1)Connect to GND; the module has a reserved TCS34725 port.

The INT, SCL, and SDA pins of the chip can directly control the color chip. In this mode, the module's MCU does not control the color chip.

Users can also set and read LED brightness levels via serial port or IIC control module. In this mode, customers should manually configure the settings online.

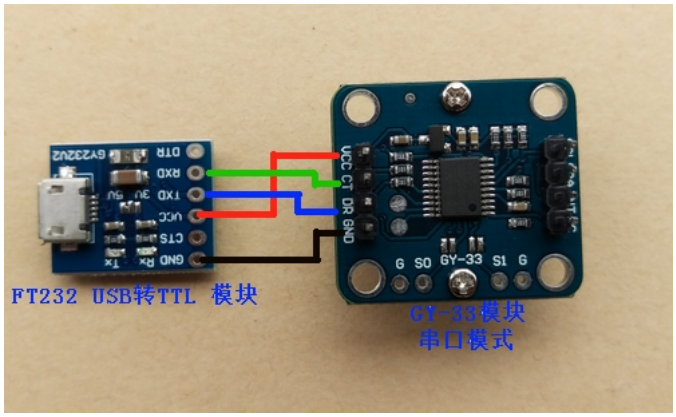
Information and programs for the TCS34725 chip can be found online; they will not be provided here.

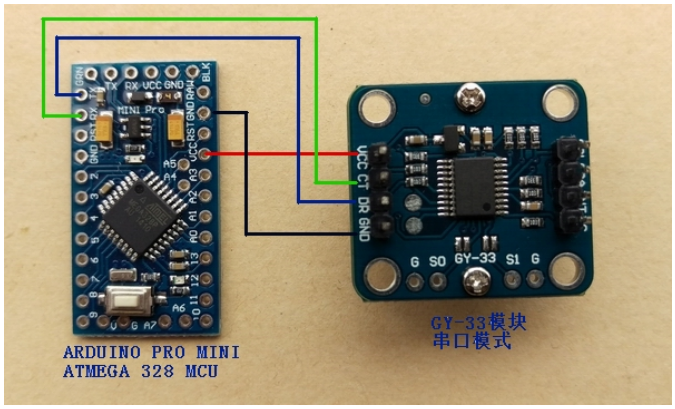
The mode selection is shown in the image below.**3kind**:

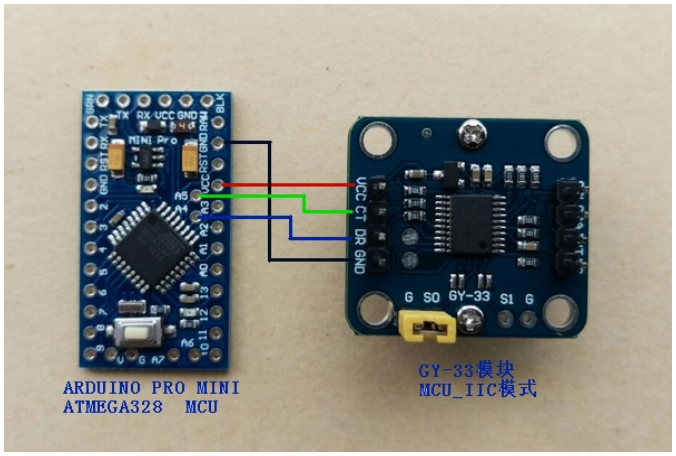


1Serial port working mode (default)2,MCU_IICmodel 3,chipIICmodel

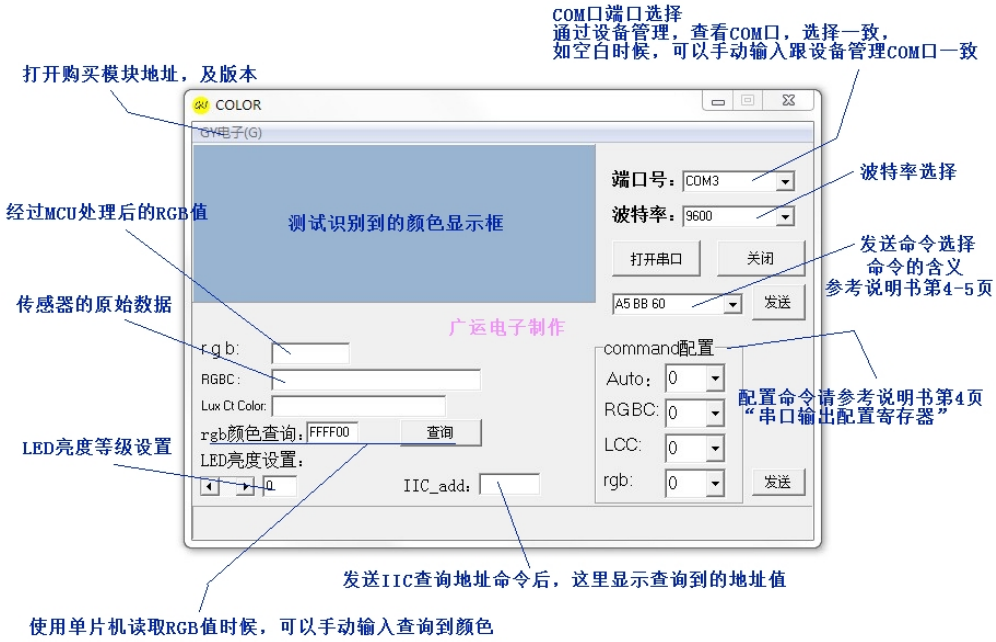
Modules andMCUorUSBchangettlConnection diagram:







Screenshots of the host computer usage:



VII. Conclusion

ModuleI/OyesTTLThe voltage level can be directly connected to the microcontroller's serial port. PL2303, CH340, FT232It can be connected to the chip, but cannot be directly connected to the computer's nine-pin serial port.

Subsequent description:

GY-33The color sensor is a full-color color detector, including a...STofMCUpieceTAOS TCS4725RGB Sensor chip and2A white lightledlamp,GY-33It can detect and measure almost all visible light within a certain range. It is suitable for colorimeter measurement applications such as color printing, medical diagnostics, computer color monitor calibration, and process control in paints, textiles, cosmetics, and printing materials.

The colors we usually see are actually the result of the object's surface absorbing some of the colored components of white light (sunlight) that shines on it, and reflecting the remaining colored light as perceived by the human eye. White light is composed of a mixture of visible light of various frequencies; that is, white light contains various colors of light (such as red, yellow, and black).R,yellowY,greenG,greenV,blueB,purpleP)According to Germany

physicist HelmholtzHelinholtz)According to the theory of the three primary colors, all colors are formed by mixing the three primary colors (red, green, and blue) in different proportions.

As can be seen from the principle of primary color sensing above, if we know the values of the three primary colors that make up all colors, we can determine the color of the object being tested.GY-33In general, when a color filter is selected, it only allows a specific primary color to pass through, blocking other primary colors. For example, when a red filter is selected, only red light can pass through, while blue and green are blocked, thus obtaining the intensity of red light. Similarly, selecting other filters will yield the intensities of blue and green light. These three intensity values can then be used to analyze the color of the light reflected onto the sensor chip.